

Manya Jain

Tempe, AZ | manyajainrkm@gmail.com | +1 480-637-1065 | linkedin.com/in/manya-jainn | github.com/ManyaJainrkm

Education

Arizona State University

Master of Science in Computer Science (GPA: 3.50/4.00)

Tempe, AZ

Expected May 2027

Manipal University Jaipur

Bachelor of Technology in Computer Engineering

India

Aug 2021 – May 2025

Skills

Languages: Python, Java, TypeScript, JavaScript, C++, C, SQL, Bash

Backend & Web: Node.js, React, TanStack Start, REST APIs, LLM streaming APIs, JSON Schema

DevOps & Cloud: Docker, K8s (K3s/K3d), Jenkins, CI/CD, Maven, WildFly, OAuth2-Proxy, Prometheus, Grafana, Linux, Git

Databases: Snowflake, PostgreSQL, MySQL, Neo4j, Redis

AI/ML: LLM evaluation, RAG, NLP, agentic workflows, sklearn, XGBoost, LightGBM, BiLSTM, TF-IDF, K-means, SHAP, LIME

Data & Scientific Computing: pandas, NumPy, SciPy, statistical testing, fairness analysis, Tableau

Experience

CIPS-AI Lab, Arizona State University

Student Researcher, Applied AI Systems

Tempe, AZ

Feb 2026 – Present

- Engineered Python ETL pipelines for legal-document ingestion, parsing, cleaning, and structural normalization, converting heterogeneous unstructured documents into consistent datasets for downstream ML classification and retrieval
- Implemented phrase-level feature extraction using bigrams and higher-order n-grams to preserve domain-specific legal terminology, capturing semantic signals not reliably represented by isolated tokens
- Modeled legal entities and relationships in Neo4j, representing cases, firms, statutes, and metadata as connected graph structures, and developed Cypher queries supporting relationship-aware retrieval and RAG workflows

Safexpress Private Limited

Software Engineer Intern, GenAI

Gurugram, India

May 2026 – Aug 2026

- Engineered configuration-driven agent logic across four Snowflake Cortex text-to-SQL skills (Orchestration, Response, Sales, Geo-Disambiguation) deployed through Slack, shipping 15+ documented fixes over six weeks
- Diagnosed agent failures through systematic functional and regression testing, tracing geo-disambiguation errors, omitted waybill filters, and output-contract violations to specific skill behaviors, then implemented targeted fixes and maintained prompt-to-fix regression records
- Designed a reusable seven-script Python analytics pipeline using pandas, scikit-learn, and SciPy to evaluate agent capability coverage, transforming 29,000+ raw interaction rows into 3,257 unique questions across 1,062 conversation threads
- Applied TF-IDF vectorization and K-means clustering to group 3,000+ questions into recurring intent types, computing a frequency-weighted skill coverage metric of 77.5% validated with a one-sample t-test ($p < 0.001$)
- Developed independent Snowflake SQL validation queries using CTEs, window functions, and anti-joins to cross-check agent-generated logic against reference behavior, producing a frequency-ranked capability gap analysis to prioritize engineering fixes

National Institute of Urban Affairs, Govt. of India

Software Engineering Intern, DevOps & Platform Systems

New Delhi, India

Aug 2024 – May 2025

- Engineered and maintained Jenkins CI/CD pipelines for 12+ containerized services, automating build validation, configuration checks, and deployment gating to improve release reliability by 30%
- Managed containerized deployments across Docker and Kubernetes (K3s/K3d), supporting repeatable build-to-deployment workflows, and implemented Prometheus metrics and Grafana dashboards for real-time service health and runtime visibility
- Supported Java-based services using Maven and WildFly, troubleshooting application startup, configuration, dependency, and runtime failures across local and containerized environments
- Diagnosed deployment and runtime failures by tracing Linux logs, container states, and OAuth2-Proxy authentication flows, reducing incident recovery time by 25% through systematic root-cause isolation

Projects

Sage: Verified AI Career-Gap Analyzer

TypeScript, React, TanStack Start, Node.js, OpenAI Responses API, JSON Schema, pdf.js

Jul 2026

- Built and deployed an end-to-end TypeScript and React application that parses resumes and arbitrary job descriptions, invokes the OpenAI Responses API, and generates structured fit assessments, prioritized skill gaps, and evidence-backed findings
- Engineered a multi-stage inference pipeline spanning browser-side PDF parsing, server-side model inference, JSON Schema-constrained outputs, and retry and fallback handling for malformed responses
- Implemented deterministic validation for resume citations and recurring skill-gap claims, rejecting unsupported evidence and surfacing degraded results rather than returning unvalidated model output

Harmoneyes: Gesture-Controlled Web Instrument

JavaScript, MediaPipe Hands, Computer Vision, Tone.js, Canvas, GitHub Pages

Jul 2026

- Built and deployed a real-time browser instrument using MediaPipe Hands, mapping webcam-derived hand landmarks to gesture states and Tone.js chord events, integrating computer-vision inference, interaction handling, and client-side audio synthesis in under 7 hours
- Implemented a gesture-to-music interaction layer translating detected hand configurations into deterministic chord mappings, rendered through Canvas-based browser visualization, with a dual-notation system connecting Western chord labels to Hindustani Sargam

Setu: Client-Side AI Support Platform

JavaScript, HTML/CSS, Streaming LLM APIs, JSON, GitHub Pages

Jul 2026

- Built and deployed a client-only AI application with seven methodology-specific conversational guides and real-time streaming LLM responses, routing requests directly from the browser to user-selected model APIs without an application server, with browser-side conversation state, user-controlled deletion, and client-side error handling for provider failures